



BITS Pilani
Pilani Campus

Digital Design

Prof. GSS Chalapathi,
Prof. Saurabh Gandhi,
Prof. Ramesh Sambangi

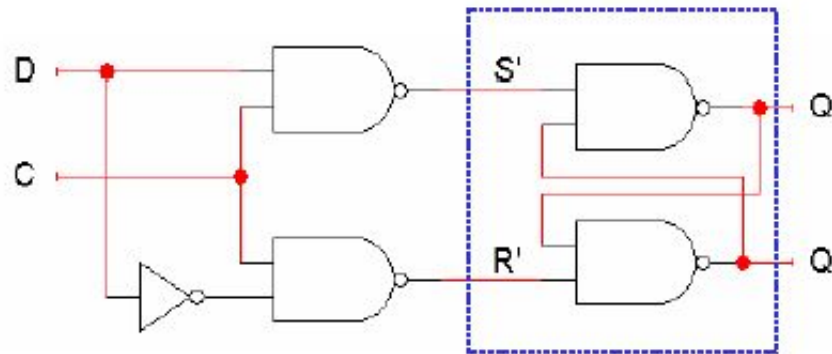
Review questions

1. In a D-latch, output becomes equal to input when...
2. In a D-flipflop, output becomes equal to input when...

The D latch (D = Data)



- A **D latch** is also based on an S'R' latch. The additional gates generate the S' and R' signals, based on inputs D ("data") and C ("control").
 - When $C = 0$, S' and R' are both 1, so Q does not change.
 - When $C = 1$, the latch output Q will equal the input D.



C	D	Q
0	x	No change
1	0	0
1	1	1

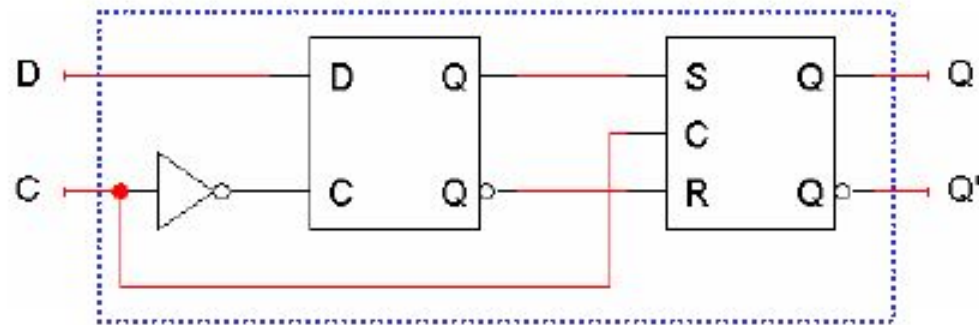
- There are two main advantages of a D latch.
 - No more messing with one input for set and another input for reset!
 - This latch has no "bad" input combinations to avoid. Any of the four possible assignments to C and D are valid.

Flip-flops

- Latches are transparent i.e. any change on inputs is immediately reflected in the output
- When control signal is high, data *continuously* flows from input to output
- Terminology: Latches are 'level sensitive'
- Synchronization across multiple memory units might become difficult and unpredictable
- Solution: 'edge sensitive' memory units, called flip-flops, that respond only at specific times

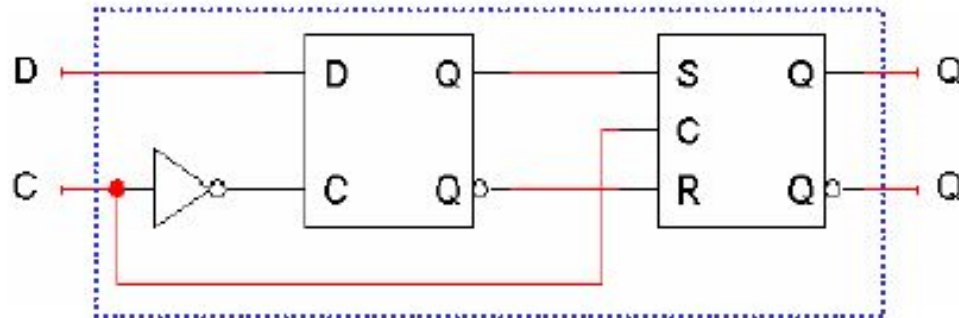
D flip-flop (D = Data or Delay...)

- Here is the internal structure of a **D-type flip-flop**.
 - The flip-flop inputs are C and D, and the outputs are Q and Q'.
 - The D latch on the left is the **master**, while the SR latch on the right is called the **slave**.



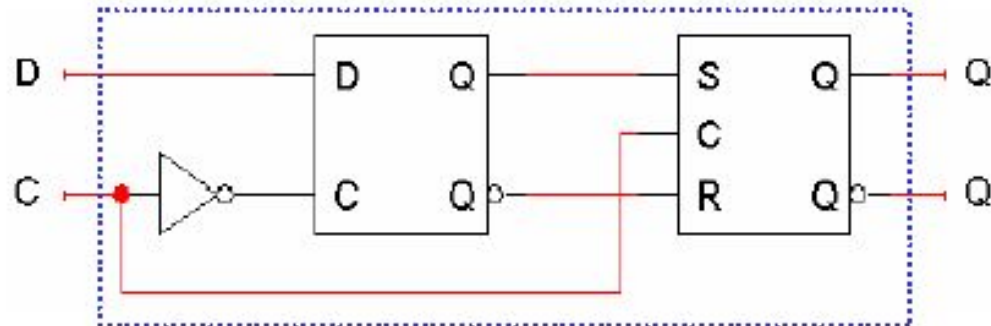
- Note the layout here.
 - The flip-flop input D is connected directly to the master latch.
 - The master latch output goes to the slave.
 - The flip-flop outputs come directly from the slave latch.

Working of D the flip-flop: $C = 0$



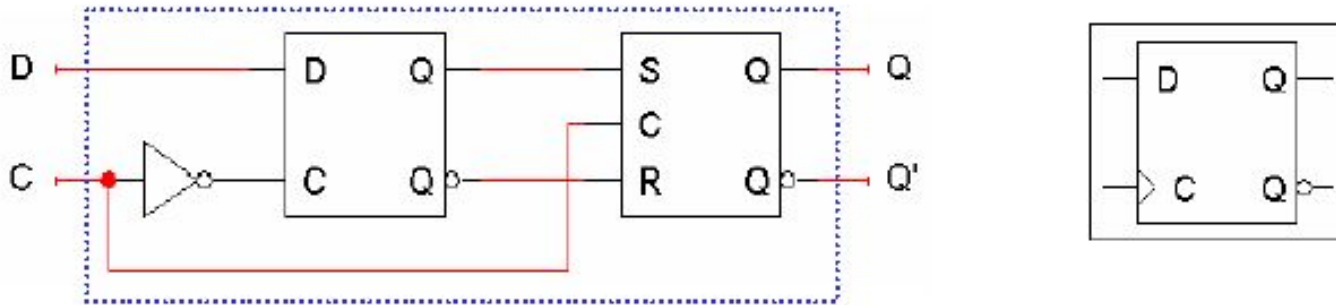
- The D flip-flop's control input C enables either the master D latch or the slave SR latch, but *not* both.
- What happens when $C = 0$?
 - The master latch is enabled, and it tracks the flip-flop input D. When D changes, the master's output changes too.
 - The slave is disabled, so the D latch output has no effect on it. Thus, the slave just maintains the flip-flop's current state.

Working of D the flip-flop: C = 1



- Several things happen *as soon as* C changes from 0 to 1.
 - The master is disabled. Its output will be the *last* D input value seen, just before C became 1. Any subsequent changes to the D input will have no effect on the master latch while C = 1.
 - On the other hand, the slave is enabled. Its output changes to reflect the master's state, which again is the D input value from right when C became 1.

Positive edge triggering



- This is called a **positive edge-triggered** flip-flop.
 - The flip-flop output Q changes *only* after the positive edge of C .
 - The change is based on the flip-flop input values that were present right at the positive edge of the clock signal.
- The D flip-flop's behavior is similar to that of a D latch, except for the positive edge-triggered nature, which is not explicit in this table.

C	D	Q
0	x	No change
1	0	0 (reset)
1	1	1 (set)

Level vs edge triggering summary



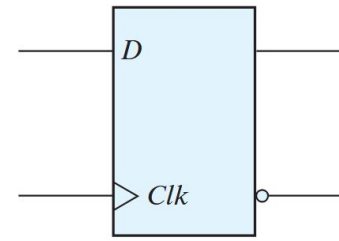
(a) Response to positive level



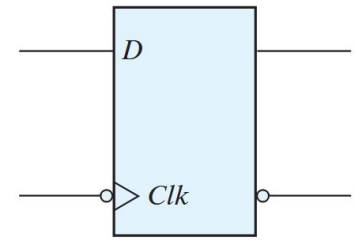
(b) Positive-edge response



(c) Negative-edge response



(a) Positive-edge



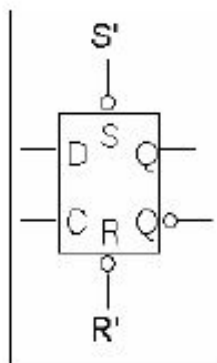
(a) Negative-edge

Fig. 5-8 Clock Response in Latch and Flip-Flop

How will you make a D flip flop that is negative edge triggered?

Flip-flops with direct inputs

- One last thing to worry about... what is the starting value of Q ?
- We could set the initial value synchronously, at the next positive clock edge, but this actually makes circuit design more difficult.
- Most flip-flops provide **direct inputs**, or **asynchronous inputs**, that let you immediately set or clear the state, regardless of the clock input.
 - You would "reset" the circuit once, to initialize the flip-flops.
 - The circuit would then begin its regular, synchronous operation.



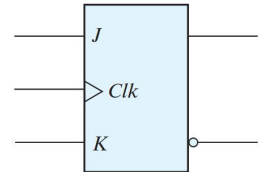
S'	R'	C	D	Q
0	0	x	x	Avoid
0	1	x	x	1 (set)
1	0	x	x	0 (reset)
1	1	0	x	No change
1	1	1	0	0 (reset)
1	1	1	1	1 (set)

} Direct inputs set or reset the flip-flop asynchronously

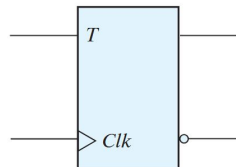
} Set $S'R' = 11$ for normal operation of the flip-flop

Other types of flip-flops

- There are three operations that can be performed with a flip-flop:
 - Set it to 1
 - Reset it to 0
 - Complement its output
- D flip-flop can set or reset the output, *no toggle*
- JK flip-flop has two inputs and performs all three operations
 - The J input sets the flip-flop to 1
 - The K input resets it to 0
 - When both inputs are enabled, the output is complemented
- T (toggle) flip-flop



(b) Graphic symbol



(c) Graphic symbol

Characteristic tables

- The tables that we've made so far are called **characteristic tables**.
 - They show the next state $Q(t+1)$ in terms of the current state $Q(t)$ and the inputs.
 - For simplicity, the control input C is usually not listed.
 - Again, these tables don't indicate the positive edge-triggered nature of the flip-flops.

D	$Q(t+1)$	Operation
0	0	Reset
1	1	Set

J	K	$Q(t+1)$	Operation
0	0	$Q(t)$	No change
0	1	0	Reset
1	0	1	Set
1	1	$Q'(t)$	Complement

T	$Q(t+1)$	Operation
0	$Q(t)$	No change
1	$Q'(t)$	Complement

Characteristic equations

- We can also write **characteristic equations**, where the next state $Q(t+1)$ is defined in terms of the current state $Q(t)$ and the flip-flop inputs.

D	$Q(t+1)$	Operation
0	0	Reset
1	1	Set

$$Q(t+1) = D$$

J	K	$Q(t+1)$	Operation
0	0	$Q(t)$	No change
0	1	0	Reset
1	0	1	Set
1	1	$Q'(t)$	Complement

$$Q(t+1) = K'Q(t) + JQ'(t)$$

T	$Q(t+1)$	Operation
0	$Q(t)$	No change
1	$Q'(t)$	Complement

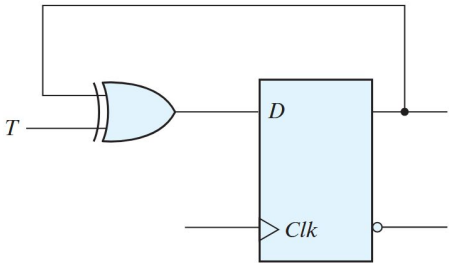
$$Q(t+1) = T'Q(t) + TQ'(t)$$

$$= T \oplus Q(t)$$

Building different flip-flops

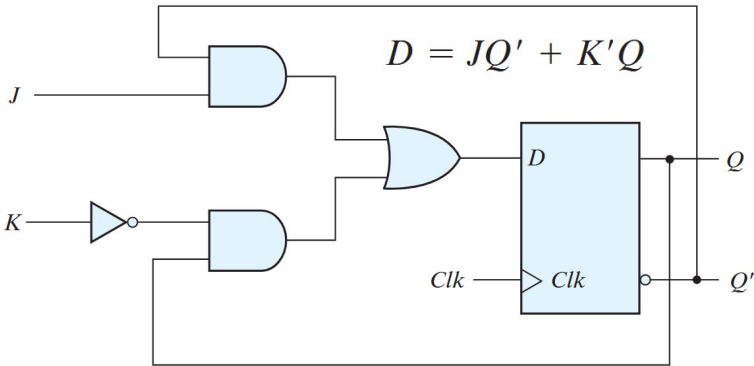
Can you build a T flip-flop from a D flip-flop?

T	Q(t+1)	Operation
0	Q(t)	No change
1	Q'(t)	Complement

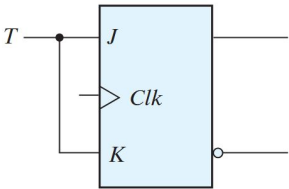


Can you build a JK flip-flop from a D flip-flop?

J	K	Q(t+1)	Operation
0	0	Q(t)	No change
0	1	0	Reset
1	0	1	Set
1	1	Q'(t)	Complement



Can you build a T flip-flop from a JK flip-flop?



Excitation tables

- Shows the minimum inputs that are necessary to generate a particular next state (in other words, to "excite" it to the next state) when the current state is known
- Rearrange the data so that the current state and next state are next to each other on the left-hand side
- Inputs needed to make that state change happen are shown on the right side of the table

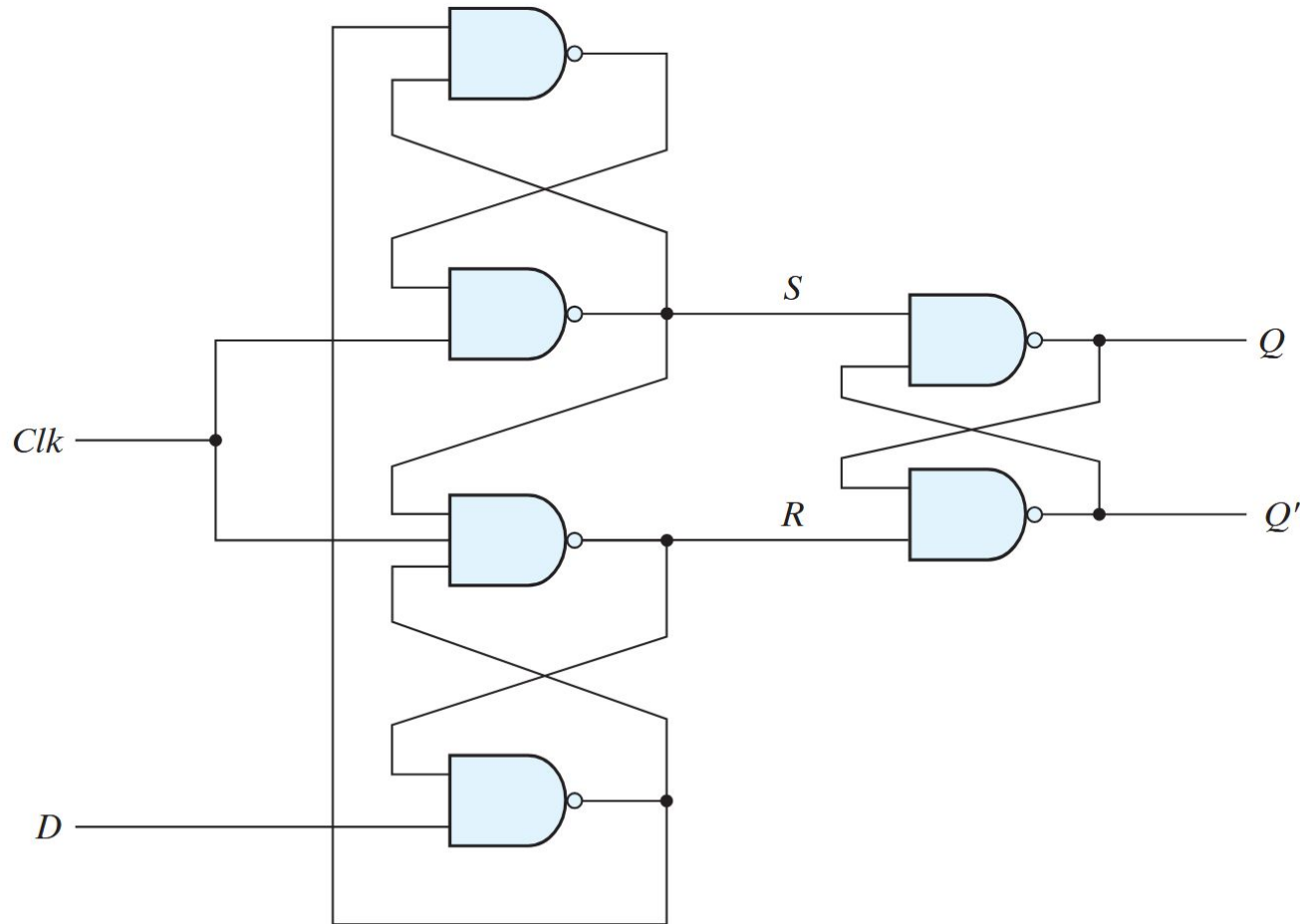
States		Input
Present	Next	D
0	0	0
0	1	1
1	0	0
1	1	1

States		Input
Present	Next	T
0	0	0
0	1	1
1	0	1
1	1	0

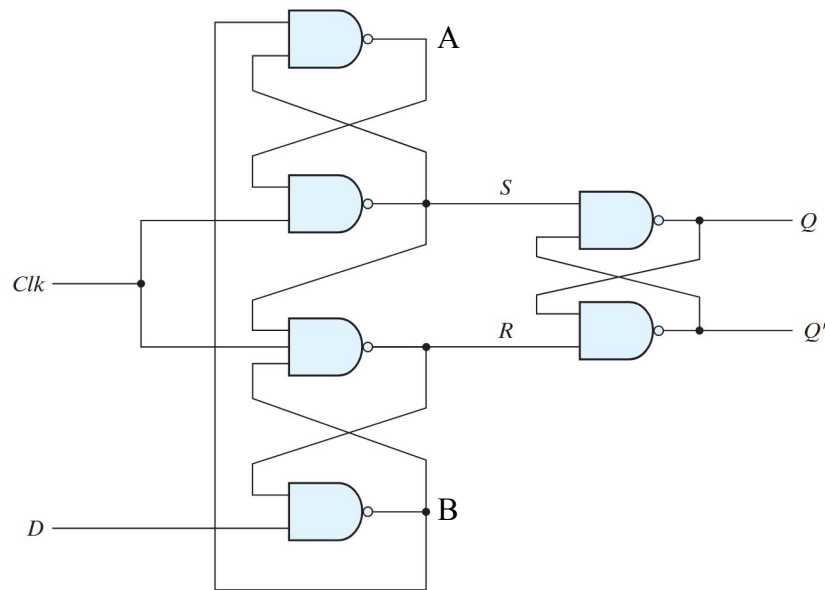
States		Inputs	
Present	Next	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

States		Inputs	
Present	Next	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

Another design of the D flip-flop



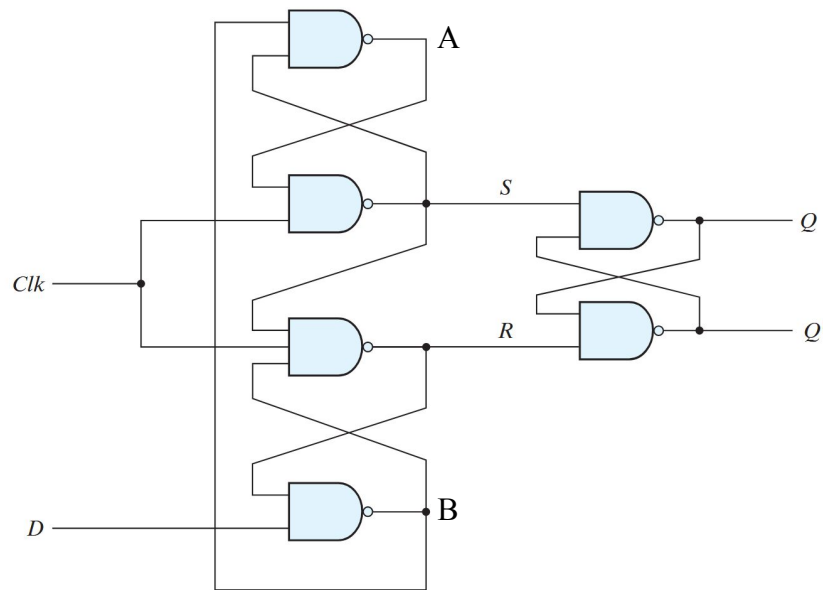
Working of the 3-latch design



Case I:

- $D = 0, CLK = 0$
- $D = 0 \Rightarrow B = 1$
- $CLK = 0 \Rightarrow R = 1, S = 1$
- $B = 1, S = 1 \Rightarrow A = 0$
- $ASRB = 0111$
- $SR = 11 \Rightarrow$ No change in Q

Working of the 3-latch design



In case I (i.e., ASRB=0111) make CLK=1

$D = 0 \Rightarrow B = 1$

- $B = 1, S = 1 \Rightarrow A = 0$

- $A = 0 \Rightarrow S = 1$

- $S = 1, B = 1, CLK = 1 \Rightarrow R = 0$

- $SR = 10 \Rightarrow \text{RESET}$

- $ASRB = 0101$

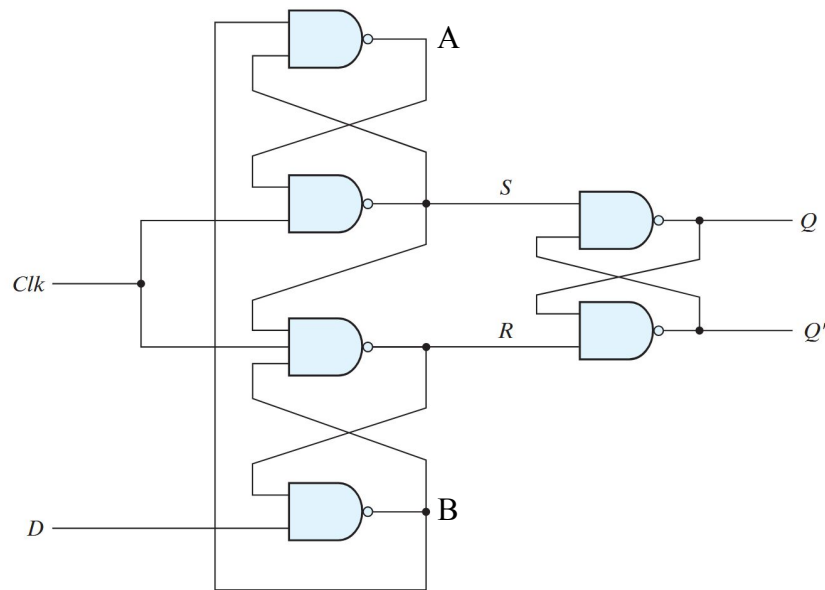
Only R has changed here after one gate delay.

What happens if D changes now:

If D changes after R has stabilized there will be no effect on output.

This is Hold Time.

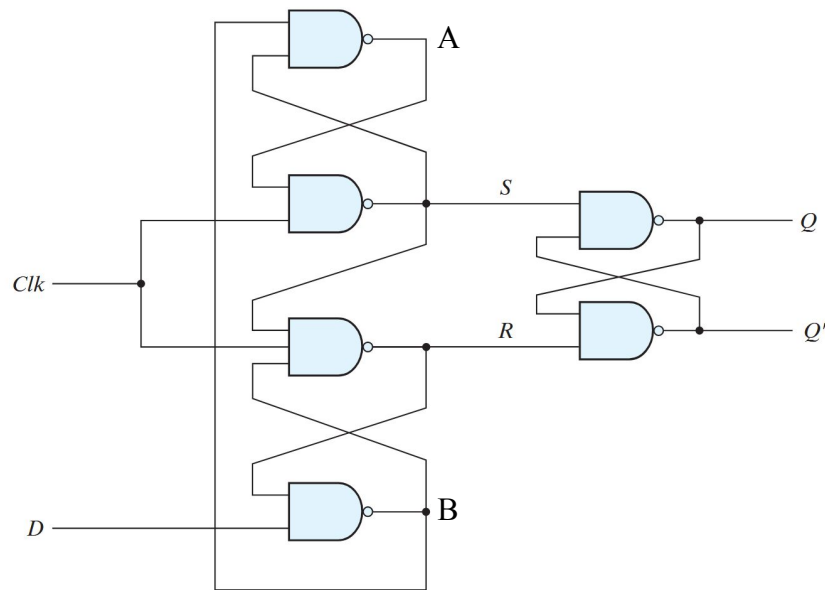
Working of the 3-latch design



Case II:

- $D = 1, CLK = 0$
- $CLK = 0 \Rightarrow R = 1, S = 1$
- $D = 1, R = 1 \Rightarrow B = 0$
- $B = 0, S = 1 \Rightarrow A = 1$
- $ASRB = 1110$
- $SR = 11 \Rightarrow \text{No change}$

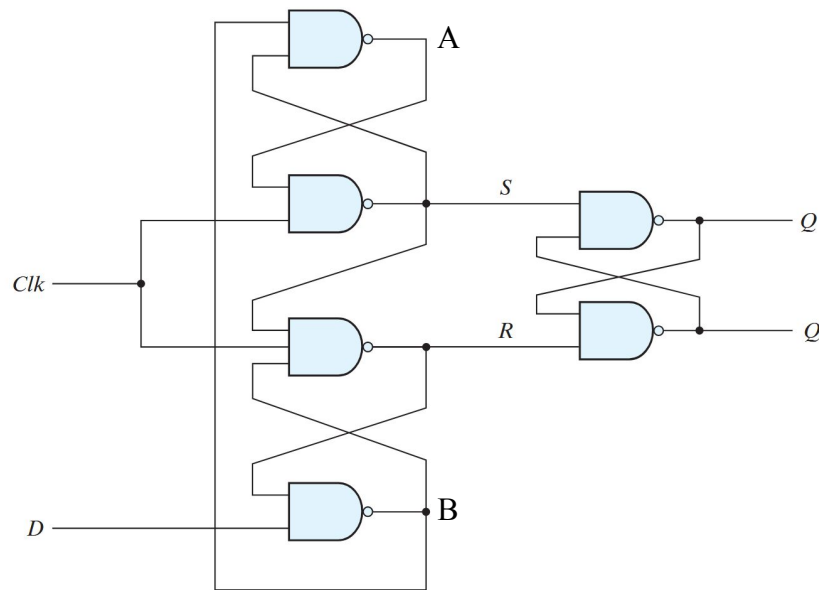
Working of the 3-latch design



In case II make $CLK = 1$

- $ASRB = 1110$
- $R = 1, D = 1 \Rightarrow B = 0$
- $B = 0, \Rightarrow A = 1$
- $A = 1, CLK = 1 \Rightarrow S = 0 \Rightarrow R = 1$
- $SR = 01 \Rightarrow SET$
- $ASRB = 1010$

Working of the 3-latch design



From $D = 1$, $CLK = 0$ what if we suddenly changed to $D = 0$, $CLK = 1$?

ASRB = 1110

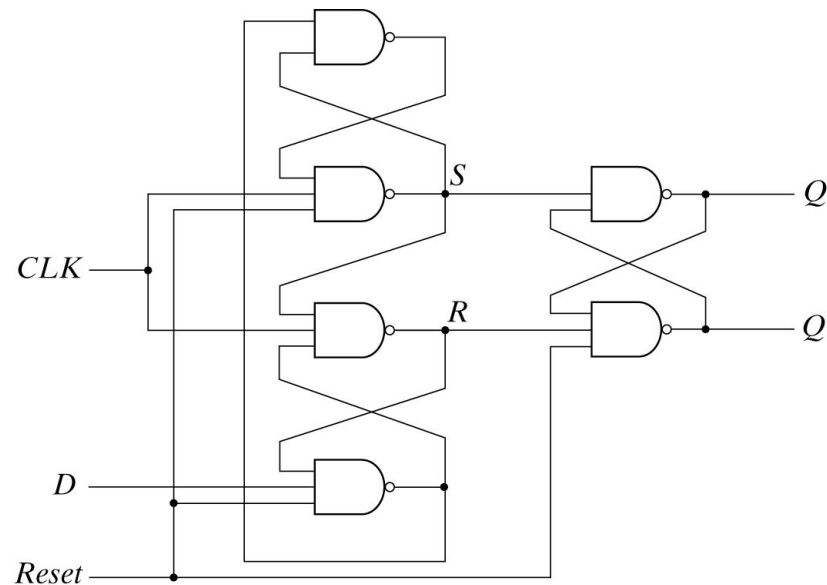
- $D = 0 \Rightarrow B = 1$
- $CLK = 1, A = 1 \Rightarrow S = 0$
- $S = 0, B = 1 \Rightarrow A = 1$
- $S = 0, B = 1, CLK = 1 \Rightarrow R = 1$
- $SR = 01 \Rightarrow SET$
- ASRB = 1011

Wrong result

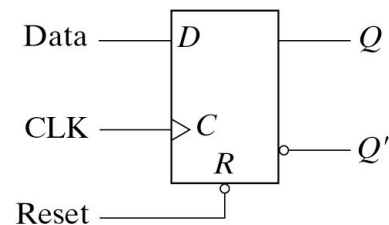
Correct sequence will be:

1. Go from case II to case I, i.e., let ASRB change from 1110 to 0111. Here A takes two gate delays to change. This is **Setup time**.
2. Now go from case I to $D = 0$, $CLK = 1$.

D flip-flop with asynchronous reset



(a) Circuit diagram



(b) Graphic symbol

R	C	D	Q	Q'
0	X	X	0	1
1	$\uparrow$	0	0	1
1	$\uparrow$	1	1	0

(b) Function table